Cynthia Leng

Local global feature fusion for image classification

Original hypothesis and design

My HW2 model was a multilayer perceptron (MLP) that classified tabby cats, cauliflower, and computer keyboards. It flattened each 32 × 32 RGB image into 3,072 inputs, followed by hidden layers of 128 and 64 neurons with ReLU and three output scores. Although the model learned to classify the images, it showed signs of overfitting and lacked the local receptive fields of a convolutional neural network (CNN).

Rather than replacing the MLP with a CNN, I proposed two parallel branches: a CNN to extract local spatial features and the original MLP to process the full image. My hypothesis was that these representations would be complementary and that an image-dependent gate could combine them more effectively than either branch alone or a fixed average. A potential failure mode is that the MLP may learn faster than the CNN, causing the gate to favor the MLP early in training and potentially limiting the CNN's contribution.

The CNN uses two 3 × 3 convolutions (3 → 16 → 32 channels), each followed by ReLU and 2 × 2 max pooling, then global pooling and a 32 → 64 projection with ReLU. Both branches produce 64 features. The gate concatenates these features, applies a 128 → 1 linear layer and sigmoid, and computes $\mathbf{F = \alpha\, L + (1 - \alpha)\, G}$, where L is the CNN representation and G is the MLP representation. A final 64 → 3 layer produces class scores. The gate starts at $\alpha = 0.5$.

Controlled experiments

I compared four architectures: MLP only, CNN only, fixed 50/50 fusion, and learned fusion. All used the same ImageNet100 subset: 600 training, 150 validation, and 150 test images, with an equal number of images from each of the three classes. Preprocessing consisted of RGB conversion and pixel scaling to 0–1. Each run used 100 epochs, cross-entropy loss, SGD with a learning rate of 0.01, batch size 32, seed 42, and no momentum or weight decay. Shared components had identical initial weights, and a separate random generator ensured that all models saw training batches in the same order. Checkpoints were selected by lowest validation loss before test evaluation.

The initial average-pooling CNN performed poorly, motivating a follow-up hypothesis: global averaging might dilute strong localized responses that are useful for classification. I changed only global average pooling to global max pooling and reran the CNN and both fusion models. This preserved parameter counts, initialization, and training settings.

Model	Global Pooling	Selected Epoch	Val Loss	Test Loss	Test Accuracy
MLP baseline	None	45	0.8761	0.8554	63.3%

CNN	Average	100	1.0917	1.0742	52.7%
Fixed fusion	Average	93	0.8752	0.8333	**66.0%**
Learned fusion	Average	70	0.8760	0.8318	64.7%
CNN	Max	100	1.0620	1.0406	57.3%
Fixed fusion	Max	93	0.8750	0.8330	**66.0%**
Learned fusion	Max	70	0.8755	0.8308	64.7%

Results and failure analysis

Learning and convergence. The models reached different validation-loss minima despite using the same training budget. The MLP was selected at epoch 45, while both fusion variants were selected at epochs 70 or 93. Both standalone CNNs achieved their lowest validation loss at the final epoch, suggesting that additional training might improve their performance. However, the current results cannot establish whether longer training would improve generalization.

Fusion performance. Fixed fusion achieved the highest test accuracy at 66.0%, compared with 63.3% for the MLP baseline. Learned fusion achieved 64.7%, outperforming the MLP but not fixed fusion. Interestingly, learned fusion had slightly lower test loss than fixed fusion despite its lower accuracy. This indicates that optimizing a learned mixing coefficient did not produce a consistent improvement. The validation losses of the MLP and fusion models were also very similar, so the test-accuracy differences provide limited evidence of a reliable advantage.

Gate behavior and possible explanations. The learned gate was intended to adapt the mixture to individual images, but it did not outperform fixed fusion in accuracy. One possible explanation is that the MLP learned useful representations faster, encouraging the gate to rely on that branch rather than fully exploiting CNN features.

Pooling experiment. Replacing global average pooling with global max pooling improved standalone CNN test accuracy from 52.7% to 57.3%. This is consistent with the hypothesis that strong localized activations may contain useful classification information. Despite the standalone CNN improvement, both fusion models retained exactly the same test accuracy after the pooling change. This suggests that improving an individual branch does not necessarily improve the combined model. The fusion architectures may already rely primarily on the MLP, or the improved CNN representations may not provide additional information useful to the classifier.

Cost and limitations. The CNN has 7,395 parameters, the MLP has 401,795, fixed fusion has 408,995, and learned fusion has 409,124. Fusion required approximately 0.43–0.45 seconds per epoch, compared with 0.26 seconds for the MLP. Therefore, the additional parameter storage was modest. Only one random seed was tested.